

01/10/26

Signal Processing

- Recap: AM, z-transform examples

- z-transform examples

$$- x[n] = \left(\frac{1}{2}\right)^n u[n] + \left(-\frac{1}{2}\right)^n u[n]$$

$$- x[n] = \left(-\frac{1}{3}\right)^n u[n] - \left(\frac{1}{2}\right)^n u[-n-1]$$

- ROC properties: ring or disc centered at 0, DFT converges iff ROC contains unit circle,

ROC has no poles, finite duration, right-sided, left-sided, two-sided

- Stability, causality, ROC: poles at $\frac{1}{2}, -2$.

- Inverse: $X(z) = \frac{1}{(1 - \frac{1}{2}z^{-1})(1 - \frac{1}{4}z^{-1})}$; $|z| > \frac{1}{2}$

- Recap: $x_1[n] = a^n \cdot u[n]$; $x_2[n] = -a^n \cdot u[-n-1]$ (anti-causal, one-sided sequence)

$$X(z) = \frac{1}{1 - az^{-1}} ; |z| > |a| ; x_2(z) = \frac{1}{1 - az^{-1}} ; |z| < |a|$$

- $x[n] = \left(\frac{1}{2}\right)^n u[n] + \left(-\frac{1}{3}\right)^n u[n]$ (one-sided sequence, causal)

$$X(z) = \frac{1}{(1 - \frac{z^{-1}}{2})} + \frac{1}{(1 + \frac{z^{-1}}{3})} ; |z| > \left|\frac{1}{2}\right|$$

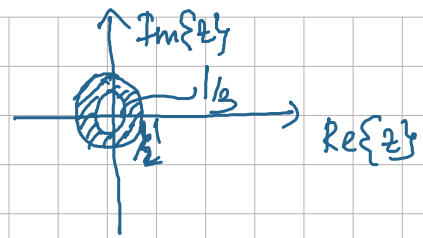
- $x[n] = \underbrace{\left(-\frac{1}{3}\right)^n u[n]}_{x_1[n]} - \underbrace{\left(\frac{1}{2}\right)^n u[-n-1]}_{x_2[n]}$ (two-sided sequence)

$$X(z) = \frac{1}{(1 + \frac{z^{-1}}{3})} - \frac{1}{(1 - \frac{z^{-1}}{2})} ; \text{ROC: } \frac{1}{3} < |z| < \frac{1}{2}$$

$$x_1(z) = \frac{1}{(1 + \frac{z^{-1}}{3})} ; |z| > \left|\frac{1}{3}\right| ; x_2(z) = \frac{1}{(1 - \frac{z^{-1}}{2})} ; |z| < \left|\frac{1}{2}\right|$$

ROC₁ ROC₂

$$ROC = ROC_1 \cap ROC_2$$



• ROC observations:

- ROC is going to be determined by circle(s) centered at 0. ($\because$ we deal with $|z|$)
- ROC cannot contain a pole. A pole is a z value for which $X(z)$ becomes infinite or does not converge. A zero is a z value where $X(z) = 0$. ($\because X(z)$ is valid only for finite values).
- The DTFT of a sequence exists if and only if the ROC of its z transform contains the unit circle. This follows from the definition of the DTFT and $z = re^{j\omega}$. i.e. $DTFT\{x[n]\} = X(z) \Big|_{|z|=1}$.
- The ROC of right-sided sequences is outside the largest pole.
- The ROC of left-sided sequences is inside the smallest pole.
- The ROC of two-sided sequences is donut shaped and lies between the smallest pole of the left sided sequence and the largest pole of the right sided sequence.
- The ROC of finite-duration sequences is going to be the entire z plane with possible exception at $z=0$ or ∞ .

Q Suppose an LTI system has a pole at $z=-2$ and $z=\frac{1}{2}$. What can we say about its ROC?

A: There are three feasible ROC.

- $ROC_1: |z| < \frac{1}{2}$
- $ROC_2: |z| > 2$

- ROC₃ : $\frac{1}{2} < |z| < 2$

Q. If $h[n]$ is causal, what is its ROC?

A: ROC₂. Since the ROC of right-sided or causal sequence is outside the largest pole.

Q. If $h[n]$ is stable, what is its ROC?

A: For stability, the DTFT must exist $\Rightarrow$ the ROC must contain the unit circle.

$\therefore$ ROC₃.

Q. If $h[n]$ is stable and causal, what is its ROC?

A: Does not exist since stability requires $\frac{1}{2} < |z| < 2$, and causality requires $|z| > 2$. These cannot be satisfied simultaneously.

• Inverse z transform:

$$X(z) = \frac{1}{(1 - \frac{1}{2}z^{-1})} \cdot \frac{1}{(1 - \frac{1}{4}z^{-1})} ; |z| > \frac{1}{2}$$

$$= \frac{2}{1 - \frac{1}{2}z^{-1}} - \frac{1}{1 - \frac{1}{4}z^{-1}}$$

By inspection we can say

$$x[n] = 2 \cdot \left(\frac{1}{2}\right)^n u[n] - \left(\frac{1}{4}\right)^n \cdot u[n] \quad (\because |z| > \frac{1}{2})$$

• Look up Prof. P. P. Vaidyanathan @ Caltech